

KMVTECH Education

Environmental & Conservation Psychology

Module 7 — Environmental Policy & Behaviour Change · Lesson 7.1

Environmental Policy & Behaviour Change**Learning Objectives**

- Define environmental policy and explain why it is necessary (externalities)
- Identify Uganda's key environmental legislation and policy bodies
- Explain the three types of policy instruments
- Apply deterrence theory, incentive-based approaches, and nudge theory
- Analyse the challenges of environmental policy implementation in Uganda

Key Terms

• Environmental policy	• Externalities
• National Environment Act Cap. 181	• NEMA
• National Climate Change Act Cap. 182	• Regulatory instruments
• Economic instruments	• Informational instruments
• Deterrence theory	• Incentive-based policy
• Nudge	• Default option
• Kaveera ban	• Polluter pays principle

1. What Is Environmental Policy and Why Is It Necessary?

Environmental policy refers to the decisions, regulations, plans, and actions governments take to manage the relationship between human activity and the natural environment.

The Externality Problem — Why Markets Fail the Environment

A factory that dumps chemical waste into a river pays nothing for the pollution — but the fishing community downstream bears the full cost. This cost imposed on others that the decision-maker does not pay is called an externality. Environmental policy exists to make decision-makers account for these costs — through taxes, regulations, and incentives that align private choices with social and environmental wellbeing.

2. Uganda's Environmental Policy Framework

Legislation / Body	What It Does
National Environment Act Cap. 181	Primary environmental legislation in Uganda — administered by NEMA
NEMA (National Environment Management Authority)	Lead body for environmental regulation, monitoring, and enforcement
National Climate Change Act Cap. 182	Legal framework for Uganda's response to climate change
National Forestry & Tree Planting Act Cap. 160	Governs forest management and protection

Water Act Cap. 164	Governs water resource management
Uganda Wildlife Act Cap. 315	Governs wildlife conservation and management
International agreements	Paris Agreement on climate change · Convention on Biological Diversity · UNCCD

3. Three Types of Policy Instruments

REGULATORY Instruments	ECONOMIC Instruments	INFORMATIONAL Instruments
• Laws, rules, standards backed by enforcement and penalties • 'Command and control' approach • Examples: banning plastic bags, requiring EIAs, prohibiting wetland encroachment • Requires enforcement to work	• Financial mechanisms making harmful behaviour costly and good behaviour rewarding • Taxes on pollution, subsidies for solar energy, payments for ecosystem services • Works through market mechanisms • Less dependent on enforcement	• Education, awareness campaigns, eco-labelling • Changes knowledge, attitudes, and values • Most people-centred instrument • Slowest to produce behaviour change

How Instruments Connect to Psychology

Regulatory instruments create the structural conditions for behaviour change (Perceived Behavioural Control in TPB). Economic instruments change the cost-benefit calculation (Attitude in TPB). Informational instruments address values, beliefs, and norms (VBN and NAM). The most effective policies use all three simultaneously.

4. How Policy Changes Behaviour — Three Mechanisms

Deterrence

Deterrence proposes that people avoid prohibited behaviour when the risk and severity of punishment outweigh the benefits. For deterrence to work two conditions must both be met:

CERTAINTY of punishment + SEVERITY of punishment → Deterrence

If either condition is weak, deterrence fails — even with severe penalties on paper

📍 Uganda Connection

NEMA is chronically underfunded relative to its mandate — insufficient field officers to monitor compliance across Uganda. This creates low certainty of punishment, which undermines deterrence even when penalties are theoretically severe. Illegal wetland encroachment continues in part because violators calculate (often correctly) that they will not be caught.

Incentive-Based Approaches

Rather than making bad behaviour costly, incentives make good behaviour rewarding. This works WITH individual self-interest rather than against it — and avoids the enforcement capacity problem.

Type of Incentive	Uganda Example
Subsidies for clean alternatives	Subsidised improved cookstoves that reduce charcoal use

Payment for Ecosystem Services (PES)	Paying farmers to maintain forest or wetland cover that protects downstream water
Tax breaks for sustainable business	Reduced taxes for businesses adopting solar energy or sustainable waste management
Social recognition	Community conservation awards that provide public recognition for environmental stewardship

Nudges — Changing Behaviour Without Changing Rules or Prices

A nudge changes the decision context to make the environmentally better choice easier, more automatic, or more socially normal — without prohibition or significant cost change.

Nudge Type	How It Works	Environmental Example
Default option	Make the sustainable choice what happens automatically unless people opt out	Make recycling the default and waste-to-landfill the opt-out
Social norm messaging	Tell people what most others are already doing	'Most vendors in this market area have switched to reusable bags'
Real-time feedback	Show people their own environmental impact compared to others	Community energy or water use data shared monthly by local leaders
Salience	Make environmental consequences visible at the point of decision	Showing an image of flooded Bwaise next to a kaveera at a market stall

5. Challenges of Environmental Policy in Uganda

Challenge	What It Means	Example
Enforcement capacity	NEMA lacks sufficient field presence to enforce compliance consistently	Wetland encroachment continues in areas where NEMA officers rarely visit
Political economy	Powerful actors with political connections influence what gets enforced	Real estate developers with political ties have built on protected wetlands without consequence
Poverty gap	Regulations that do not account for poverty push violation costs onto those with no alternative	A charcoal ban without affordable LPG alternatives criminalises poverty
Time horizon	Environmental benefits are long-term; political cycles are short	Politicians prioritise visible short-term development over invisible long-term environmental protection

🔗 Uganda Connection

The kaveera ban's trajectory shows both the challenge and the path forward. Initial compliance was weak because enforcement was inconsistent and alternatives were unavailable. Over time — as enforcement improved, alternatives became accessible, and the social norm shifted — compliance increased. The lesson: effective environmental policy combines all three instruments AND takes time. Quick wins through regulation alone rarely hold.

✔ Lesson Summary

1. Environmental policy exists because markets fail to account for externalities — costs imposed on others
2. Uganda's framework: National Environment Act Cap. 181, NEMA, National Climate Change Act Cap. 182

3. Three instrument types: regulatory (command and control), economic (incentives), informational (education)

4. Deterrence requires both certainty AND severity of punishment — Uganda's enforcement gaps reduce both

5. Nudges change decision contexts without prohibiting options — underused but cost-effective in Uganda

6. Key challenges: enforcement capacity, political economy, poverty gap, and short political time horizons

Quick Revision

1. What is an externality and why does it justify environmental policy?

2. Name Uganda's primary environmental legislation and the body that administers it.

3. Describe the three types of policy instruments and give one Uganda example of each.

4. What two conditions must be met for deterrence to work?

5. What is a nudge? Give one environmental nudge that could be used in Uganda.

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Environmental & Conservation Psychology

Module 8 — Economic Policy & Global Environmental Concerns · Lesson 8.1

Economic Policy & Global Environmental Concerns

📖 Learning Objectives

- Explain four ways economic activity affects the environment
- Evaluate the Environmental Kuznets Curve hypothesis and its limitations
- Identify six major global environmental concerns
- Explain the poverty-environment cycle in both directions
- Identify the most relevant SDGs for Uganda's environmental context

📖 Key Terms

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|-------------------------------|-----------------------------|
| • Resource extraction | • Pollution |
| • Land transformation | • Climate change |
| • Environmental Kuznets Curve | • Sustainable development |
| • Ecosystem services | • Poverty-environment cycle |
| • SDGs | • SDG 13 |
| • SDG 15 | • SDG 6 |
| • SDG 1 | • Biodiversity loss |
| • Deforestation | • Freshwater scarcity |
| • Plastic pollution | |

1. Four Ways Economic Activity Harms the Environment

RESOURCE Extraction	POLLUTION	LAND Transformation
• Mining, logging, fishing, agriculture use natural materials • When extraction exceeds regeneration rate → degradation • Uganda: oil in Albertine Rift, charcoal production, sand mining	• Production & consumption generate waste • Industrial discharge, vehicle emissions, agricultural chemicals, plastic • Uganda: Lake Victoria, Lake Albert, and the Nile under pressure	• Development converts forests, wetlands, grasslands • Destroys habitat, alters water cycles, reduces ecosystem services • Uganda: wetlands drained for housing, forests cleared for agriculture

Fourth pathway: CLIMATE CHANGE	
Fossil fuel combustion and deforestation release greenhouse gases	Drives temperature rise, extreme weather, sea level rise, agricultural disruption
Uganda contributes less than 0.1% of global emissions	But is among Africa's most climate-vulnerable countries

2. The Environmental Kuznets Curve (EKC) Hypothesis

The EKC proposes an inverted U-shape: as countries develop economically, environmental degradation first increases — then, beyond an income threshold, begins to improve.

LOW INCOME ↗ Degradation increasing Production prioritised, environmental costs externalised	PEAK ▲ Maximum degradation Turning point — income high enough to afford protection	HIGH INCOME ↘ Degradation decreasing Cleaner technology, environmental regulation, political will
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EKC Critique — Why 'Develop First, Clean Up Later' Is Problematic	Relevance to Uganda
Income threshold for improvement is very high — beyond most developing countries for decades	Uganda may wait generations before reaching the threshold
Environmental improvements in rich countries partly achieved by offshoring dirty production to poor ones	Uganda may become the recipient of dirty industries relocated from wealthier countries
Some damage (biodiversity loss, climate change) may be irreversible before income rises sufficiently	Uganda's forests and wetlands once lost may not recover in any relevant human time frame

3. Six Major Global Environmental Concerns

Concern	Global Picture	Uganda Dimension
Climate change	Greenhouse gas emissions driving temperature rise, extreme weather, sea level rise	Uganda contributes <0.1% of emissions but faces floods, droughts, and agricultural disruption
Biodiversity loss	Wildlife populations down ~69% since 1970 (WWF). Sixth mass extinction underway	Uganda hosts exceptional primate, bird, and plant diversity — all under pressure
Deforestation	Forests cover 31% of land, store 80% of terrestrial species	Uganda's forest cover has declined by over 50% since the 1990s
Freshwater scarcity	Only 3% of water is freshwater; growing demand from population, agriculture, industry	Uganda's water resources (Lake Victoria, Nile, wetlands) under pressure from pollution and climate change
Plastic pollution	400 million tonnes produced annually; microplastics found in human blood	Visible daily challenge in Kampala, along lake shores, and in wetlands
Land degradation	1/3 of world's land significantly degraded	Soil erosion in hilly eastern/southwestern Uganda; overgrazing in Karamoja

4. The Poverty-Environment Cycle

The poverty-environment cycle (nexus) describes a self-reinforcing relationship: poverty drives environmental degradation, and environmental degradation in turn deepens poverty.



ENVIRONMENTAL DEGRADATION Deforestation · Soil erosion · Water pollution · Wetland drainage · Overgrazing

DEEPENED POVERTY Reduced rainfall → crop failure · Reduced fish stocks → lost livelihoods · Increased flooding → home destruction · Lost medicinal plants → increased healthcare costs

How Poverty Drives Degradation	How Degradation Deepens Poverty
Cannot afford LPG → burns charcoal → deforestation	Deforestation reduces rainfall → crop failures → food insecurity
No land security → overexploits soil rather than investing in long-term management	Soil erosion reduces agricultural productivity → loss of income
No access to sanitation → uses rivers for waste → water pollution	Water pollution reduces fish stocks and increases disease burden
No political power → cannot stop polluters or developers from exploiting local environments	Flood risk increases as wetland protection disappears → repeated home destruction

Breaking the Cycle

Integrated approaches addressing BOTH poverty and environmental degradation simultaneously are most effective: sustainable livelihood programs that provide alternatives to destructive activities; community-based resource management giving communities rights AND responsibilities; ecosystem service payments compensating communities for environmental stewardship; governance reform giving poor communities a voice in decisions affecting their environment.

5. The Sustainable Development Goals (SDGs)

The 17 SDGs were adopted by all UN member states in 2015 as part of the 2030 Agenda. They explicitly link environmental protection to poverty reduction, health, and economic development — reflecting the integrated approach required to address the poverty-environment cycle.

SDG	Title	Relevance to Uganda
SDG 1	No Poverty	Foundation of addressing poverty-environment cycle
SDG 2	Zero Hunger	Agricultural sustainability and food security linked to environmental health
SDG 6	Clean Water	Lake Victoria, Nile, and Uganda's wetlands under pressure
SDG 13	Climate Action	Uganda among Africa's most climate-vulnerable despite minimal emissions
SDG 14	Life Below Water	Lake Victoria, fisheries, and aquatic biodiversity
SDG 15	Life on Land	Uganda's forests, wetlands, and exceptional terrestrial biodiversity

?? Uganda Connection

Three Uganda cases illustrate economy-environment tensions: (1) Oil in the Albertine Rift — enormous revenue potential vs habitat destruction and community displacement. (2) Agricultural expansion — feeds families but drives deforestation and wetland loss. (3) Charcoal energy poverty — cooking fuel for millions but drives tree cover loss. In all three, the most important insight is the same: economic wellbeing in the long run **DEPENDS** on the health of the environmental systems being degraded.

✓ Lesson Summary

1. Economic activity harms the environment through resource extraction, pollution, land transformation, and climate change
2. EKC hypothesis argues degradation rises then falls with income — but this is contested and risky for Uganda
3. Six global concerns: climate change, biodiversity loss, deforestation, freshwater scarcity, plastic pollution, land degradation
4. Poverty-environment cycle is self-reinforcing — poverty drives degradation and degradation deepens poverty
5. Breaking the cycle requires integrated approaches addressing both poverty and environment simultaneously
6. SDGs 1, 6, 13, 14, and 15 are most directly relevant to Uganda's environmental context

📝 Quick Revision

1. Name the four ways economic activity harms the environment.
2. What does the Environmental Kuznets Curve predict, and why is it problematic for Uganda?
3. Name four of the six major global environmental concerns.
4. Explain the poverty-environment cycle in BOTH directions with one Uganda example for each direction.
5. Which SDGs are most relevant to Uganda and why?

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Environmental & Conservation Psychology
 Module 9 — Applying Environmental Psychology · Lesson 9.1
Careers, Applications & Next Steps

Learning Objectives

- Identify six career pathways for environmental psychology graduates in Uganda
- Explain how environmental psychology knowledge applies across different fields
- Connect course concepts to real-world environmental challenges in Uganda
- Outline practical personal next steps for applying what you have learned

Key Terms

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| • Environmental NGO careers | • NEMA careers |
| • Environmental journalism | • Urban planning psychology |
| • International development | • Community development |
| • NAPE Uganda | • WWF Uganda |
| • Uganda Radio Network | • Makerere University |

1. Six Career Pathways in Uganda & Beyond

Career Pathway	What Environmental Psychology Offers	Key Organisations
NGO & Civil Society	Design behaviour change programs; understand community motivation; evaluate campaign effectiveness	NAPE, WWF Uganda, Nature Uganda, Jane Goodall Institute Uganda, AFIEGO, Uganda Conservation Foundation
Government Environmental Management	Design public awareness campaigns; integrate community participation into EIA processes; advise on compliance strategy	NEMA, Ministry of Water & Environment, Ministry of Agriculture, Local Governments
Environmental Journalism & Communication	Frame messages for specific audiences; counter misinformation; understand psychological effects of coverage	Daily Monitor, New Vision, Uganda Radio Network, NBS TV, Stanbic Media Foundation
Research & Academia	Generate Uganda-specific environmental psychology research; evaluate program effectiveness; contribute to global literature	Makerere University, Uganda Christian University, Uganda Management Institute
Urban Planning & Community Development	Apply knowledge of green space, crowding, noise, place identity to housing and urban design decisions	Kampala Capital City Authority, National Planning Authority, development consultancies
International Development & Climate Finance	Work across technical environmental knowledge and social science of behaviour change	UNDP, UNEP, UN-Habitat, GIZ, USAID, World Bank, African Development Bank

2. How This Course Applies Across Fields

You do not need to work in an 'environmental' job for this course to be directly useful.

Your Field	How Environmental Psychology Applies
Law	Understand why communities resist resettlement; apply place identity in land rights cases; use NAM and TPB to advise on compliance design
Health	Recognise eco-anxiety and climate grief in patients; apply disaster psychology in health responses; design health facilities using ART and SRT principles
Journalism	Frame environmental stories for maximum impact; identify and counter misinformation; understand what makes communities engage with environmental news
Business	Design supply chains that address environmental attitudes; understand consumer pro-environmental behaviour; develop CSR programs that activate rather than bypass values
Teaching	Build environmental awareness that activates moral responsibility not just knowledge; connect environmental issues to students' own community experience
Community Development	Apply participation ladder principles; use Place Identity and NAM to design community conservation programs that actually sustain themselves

3. What to Do With What You Have Learned

1. See your environment differently

When you see a wetland being encroached upon, you now understand the values, attitudes, norms, and economic pressures driving it. When you see a campaign failing, you can identify which psychological barriers are not being addressed. This is practical intelligence.

2. Apply frameworks in your specific field

Do not leave TPB, NAM, VBN, ART, and the others in the classroom. Every field represented in this course has a place for these frameworks in real daily work.

3. Engage your community

You now understand the conditions for effective community engagement: building place identity and connection, activating moral responsibility, working through trusted leaders, removing structural barriers alongside psychological ones.

4. Continue learning

Environmental psychology is young and evolving fast — particularly research on eco-anxiety, climate grief, and climate action psychology. The challenges that make this field urgent are not going away. Stay curious.

5. Act

Research consistently finds that taking meaningful action — however small — is the most effective antidote to eco-anxiety and helplessness. Work within your sphere of influence.

A Closing Thought

Uganda is among the countries most severely affected by climate change despite contributing less than 0.1% of global greenhouse gas emissions. The people most harmed by a global problem they did not cause are Ugandans. Environmental psychology tells us that solving this requires understanding human behaviour — the values behind it, the barriers to changing it, the social norms that sustain it, the psychological costs of living with environmental destruction. That is what this course has given you. Use it.

✓ Lesson Summary

1. Six career pathways: NGO work, government environmental management, journalism, research, urban planning, international development

2. Environmental psychology knowledge is applicable across law, health, journalism, business, teaching, and community development

3. Seeing your environment through psychological frameworks is itself a practical professional skill

4. The most effective antidote to eco-anxiety is meaningful action within your sphere of influence

5. Uganda's environmental challenges are urgent, real, and require psychologically informed approaches — not just technical solutions

Quick Revision

1. Name four career sectors in Uganda where environmental psychology knowledge is directly applicable.

2. How could a journalist use knowledge from this course to produce more effective environmental stories?

3. Why is environmental psychology particularly relevant to Uganda's development context?

4. What does research say is the most effective way to manage eco-anxiety?

5. Explain in one sentence the most important thing this course has given you.

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